# COEFFICIENTWISE LOG-CONCAVITY OF THE CUBIC MULTIPLE-POCHHAMMER SERIES

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Abstract. Let $(f_n)_{n\geq 1}$ be a nonnegative log-concave sequence with no internal zeros, and define the cubic multiple-Pochhammer series $F_f(\mu;x) = \sum_{n\geq 1} f_n\,(\mu)_{3n}\,x^n/(3n-1)!$. Karp and Zhang established coefficientwise log-concavity of the analogous series for multiplicity $r=2$ and gave numerical counterexamples for $r=4$, leaving the cubic case open as their Conjecture 1. We prove it: for all $\mu,\alpha,\beta\geq 0$, every coefficient of the generalized Turánian $F_f(\mu+\alpha;x)F_f(\mu+\beta;x) - F_f(\mu;x)F_f(\mu+\alpha+\beta;x)$ is nonnegative. Equivalently, the map $(f_n)\mapsto F_f$ carries every such sequence to a coefficientwise log-concave family. The proof is elementary and noncomputational. The degree-$m$ coefficient of each product $F_f(u;x)F_f(v;x)$ is represented as a beta-binomial expectation, reducing the theorem to the monotonicity of a weighted residue-class binomial kernel. The constant-weight case rests on a Bernstein-basis positivity certificate obtained from a root-of-unity filter; monotone weights are then inserted through a one-sign-change decomposition of the complementary residue blocks. A coefficientwise strictness criterion follows, as does ordinary log-concavity in $\mu$ on $(0,\infty)$ for $(f_n)\not\equiv 0$ at every positive $x$ within the radius of convergence.

## 1. Introduction

A formal power series depending on a parameter $\mu\geq 0$ is called *coefficientwise log-concave* if, for every $\mu,\alpha,\beta\geq 0$, all coefficients of the generalized Turánian

$$F(\mu+\alpha;x)F(\mu+\beta;x) - F(\mu;x)F(\mu+\alpha+\beta;x)$$

are nonnegative. This property is stronger than the corresponding pointwise inequality: whenever the relevant series converge at a positive value of $x$, summing the coefficient inequalities yields the generalized Turán inequality for $\mu\mapsto F(\mu;x)$ there, simultaneously for all admissible shifts. It has been studied systematically for reciprocal-gamma, gamma-ratio, hypergeometric, and Fox–Wright series; see [4, 5, 6, 7]. Turán-type inequalities for hypergeometric and confluent hypergeometric functions have also been studied extensively [1, 2].

Karp and Zhang considered, for an integer $r\geq 2$, the modified multiple-Pochhammer series

$$F_{f,r}(\mu;x) = \sum_{n\geq 1} f_n \frac{(\mu)_{rn}}{(rn-1)!}x^n, \tag{1.1}$$

where $(f_n)$ is nonnegative and log-concave. Karp and Zhang report that A. Ninh suggested this construction to them in a private communication in 2017 [7]; related published work on log-concavity and Turán-type inequalities appears in [9]. They proved coefficientwise log-concavity for $r=2$ [7, Thm. 2] and gave numerical counterexamples for $r=4$ [7, Rem. 5], leaving the cubic case $r=3$ open as their Conjecture 1. This note proves it.

Recall that a nonnegative sequence $(f_n)_{n\geq 1}$ is log-concave with no internal zeros if $f_n^2\geq f_{n-1}f_{n+1}$ whenever the three terms are defined, and its positive support is an interval of integers. A nontrivial such sequence is a Pólya frequency sequence of order two, or $\mathrm{PF}_2$, also called doubly

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positive. The interval-support condition does not follow from the inequality, and Theorem 1.1 fails without it (Remark 5.4).

Karp and Zhang build the inequality and the interval-support condition into their definition of log-concavity [7, Def. 1]. They index the ambient sequence from $n = 0$, although the modified series itself begins at $n = 1$. The log-concavity hypothesis is stated here directly for the positive-index tail $(f_n)_{n\geq 1}$. The two formulations are equivalent: restriction preserves log-concavity and interval support, while any admissible tail extends to $n = 0$ by setting $f_0 = 0$. The hypotheses of Theorem 1.1 are therefore those of their Conjecture 1.

Except in Corollary 5.3, we work with formal power series, so no growth condition on $(f_n)$ is needed.

**Theorem 1.1** (Cubic multiple-Pochhammer theorem)**.** *Let $(f_n)_{n\geq 1}$ be a nonnegative log-concave sequence with no internal zeros, and put*

$$F_f(\mu; x) = \sum_{n\geq 1} f_n \frac{(\mu)_{3n}}{(3n-1)!} x^n. \tag{1.2}$$

*For every $\mu, \alpha, \beta \geq 0$, the formal power series*

$$\Delta_f(\mu; \alpha, \beta; x) = F_f(\mu + \alpha; x) F_f(\mu + \beta; x) - F_f(\mu; x) F_f(\mu + \alpha + \beta; x) \tag{1.3}$$

*has nonnegative coefficients.*

Theorem 1.1 concerns the transform $(f_n) \mapsto F_f$, not a single function: every nonnegative log-concave sequence is sent to a parameter family whose generalized Turánians have nonnegative coefficients. Summing at $\alpha = \beta$ and invoking continuity yields ordinary log-concavity of $\mu \mapsto F_f(\mu; x)$ where the series converges (Corollary 5.3). The cubic case is the intervening multiplicity between the positive case $r = 2$ and the first known failure at $r = 4$, and the residue-class positivity of Section 4 is the mechanism behind it.

For $f_n \equiv 1$, the triplication formula for the Pochhammer symbol gives

$$F_f(\mu; x) = 3x \frac{\mathrm{d}}{\mathrm{d}x} {}_3F_2\left(\tfrac{\mu}{3}, \tfrac{\mu+1}{3}, \tfrac{\mu+2}{3}; \tfrac{1}{3}, \tfrac{2}{3}; x\right),$$

so the theorem yields coefficientwise generalized Turán inequalities for the differentiated ${}_3F_2$ family above; a general log-concave $(f_n)$ gives a coefficient-weighted extension.

The proof has three components. The degree-$m$ coefficient of each product $F_f(u; x)F_f(v; x)$ is rewritten as a beta-binomial expectation (Section 2). Two elementary monotonicity lemmas (Section 3) then supply the tools: beta laws with a fixed sum of parameters are ordered by their imbalance against symmetric functions increasing toward the midpoint, and a one-sign-change weighting principle carries positivity through nondecreasing weights. The weighted residue-class binomial kernel is shown to have the required monotonicity (Section 4); here complementary residue blocks have a single sign change, and the unweighted sum carries a Bernstein-basis positivity certificate. This kernel theorem does not depend on the beta-binomial setup. The components are combined in Section 5.

## 2. Beta-binomial representation

This section rewrites the coefficient convolution underlying the Turánian (1.3) as a beta-binomial expectation.

For $m \geq 2$, put

$$w_k = f_k f_{m-k}, \qquad 1 \leq k \leq m - 1. \tag{2.1}$$

Log-concavity of $(f_n)$ implies

$$w_{k+1} = f_{k+1} f_{m-k-1} \geq f_k f_{m-k} = w_k \qquad \left(1 \leq k < \left\lfloor \tfrac{m}{2} \right\rfloor\right),$$

the standard cross-product consequence $f_{a+1} f_b \geq f_a f_{b+1}$ for $a < b$ of log-concavity with no internal zeros. With the symmetry $w_k = w_{m-k}$, the weights (2.1) satisfy the hypotheses of Theorem 4.1.

For $u, v \geq 0$, the degree-$m$ coefficient of the product $F_f(u;x)F_f(v;x)$ is

$$C_{m,f}(u,v) = \sum_{k=1}^{m-1} w_k \frac{(u)_{3k}(v)_{3(m-k)}}{(3k-1)![3(m-k)-1]!}, \tag{2.2}$$

with $C_{m,f}(0,v) = 0$ since $(0)_{3k} = 0$. The beta-binomial representation below is used for $u, v > 0$. Let $N = 3m$ and let $J$ have the beta-binomial distribution

$$\mathbb{P}(J = j) = \binom{N}{j} \frac{(u)_j (v)_{N-j}}{(u+v)_N}, \qquad 0 \leq j \leq N.$$

For $1 \leq j \leq N-1$,

$$\frac{(u)_j (v)_{N-j}}{(j-1)!(N-j-1)!} = \frac{(u+v)_N}{N!} j(N-j)\mathbb{P}(J = j).$$

Defining a weight on all integers $0 \leq j \leq N$ by

$$\widetilde{w}_j = \begin{cases} w_{j/3}, & 0 < j < N \text{ and } 3 \mid j, \\ 0, & \text{otherwise,} \end{cases}$$

we obtain

$$C_{m,f}(u,v) = \frac{(u+v)_N}{N!} \mathbb{E}[J(N-J)\widetilde{w}_J]\,. \tag{2.3}$$

The factor $J(N-J)$ vanishes at $j = 0$ and $j = N$, so the endpoint terms do not contribute.

Use the standard mixture representation $P \sim \mathrm{Beta}(u,v)$ and $J \mid P \sim \mathrm{Bin}(N,P)$. The identity

$$j(N-j)\binom{N}{j} = N(N-1)\binom{N-2}{j-1}$$

then transforms (2.3) into

$$C_{m,f}(u,v) = \frac{(u+v)_N}{N!} N(N-1)\mathbb{E}\, G_{m,w}(P), \tag{2.4}$$

where the kernel

$$G_{m,w}(p) = \sum_{k=1}^{m-1} w_k \binom{3m-2}{3k-1} p^{3k}(1-p)^{3(m-k)}, \qquad 0 \leq p \leq 1, \tag{2.5}$$

is defined for arbitrary symmetric nonnegative weights $w_k = w_{m-k}$ and is symmetric under $p \mapsto 1-p$. The congruence $j \equiv 0 \pmod 3$ becomes $j - 1 \equiv 2 \pmod 3$, which explains the binomial coefficient $\binom{3m-2}{3k-1}$.

By (2.4) and Lemma 3.2, Theorem 1.1 reduces to the monotonicity of $G_{m,w}$ on $[0, 1/2]$ (Section 4).

## 3. Two monotonicity lemmas

This section records two elementary monotonicity facts. The weighting principle for the paired residue blocks of Section 4 is due to [5, Lem. 6], proved in [4, Lem. 2.1], and was applied to multiplicity two in [7]. The ordering of beta laws by imbalance used in Section 5 is a fixed-sum likelihood-ratio comparison for folded beta laws, obtained in [3, Thm. 3] under the restriction that each parameter gap be 0 or at least 1, and proved below without that restriction.

**Lemma 3.1.** *Let $0 \leq w_1 \leq w_2 \leq \cdots \leq w_L$ and suppose the real sequence $A_1, \ldots, A_L$ has no more than one sign change, from nonpositive to nonnegative. If*

$$\sum_{k=1}^{L} A_k \geq 0,$$

*then*

$$\sum_{k=1}^{L} w_k A_k \geq 0.$$

*If in addition* $\sum_{k=1}^{L} A_k > 0$*,* $w_L > 0$*, and* $A_L > 0$*, then* $\sum_{k=1}^{L} w_k A_k > 0$*.*

*Proof.* The claim is immediate if all $A_k$ are nonnegative. Otherwise, let $k_0$ be the largest index for which $A_{k_0} < 0$. Put $c = w_{k_0}$. For $k \le k_0$ we have $w_k \le c$ and $A_k \le 0$, while for $k > k_0$ we have $w_k \ge c$ and $A_k \ge 0$. Therefore

$$\sum_{k=1}^{L} w_k A_k \ge c \sum_{k=1}^{L} A_k \ge 0.$$

For the strict statement, assume $\sum_k A_k > 0$, $w_L > 0$, and $A_L > 0$. If every $A_k$ is nonnegative, then $\sum_k w_k A_k \ge w_L A_L > 0$. Otherwise $A_L > 0$ forces $k_0 < L$. If $c > 0$, then $\sum_k w_k A_k \ge c \sum_k A_k > 0$. If $c = 0$, then $w_k = 0$ for all $k \le k_0$, so $\sum_k w_k A_k = \sum_{k>k_0} w_k A_k \ge w_L A_L > 0$. □

**Lemma 3.2.** *Fix* $s > 0$*. For* $0 \le d < s/2$*, let* $P_d \sim \mathrm{Beta}(s/2 + d, s/2 - d)$*. If* $H : [0, 1] \to \mathbb{R}$ *is symmetric,* $H(p) = H(1 - p)$*, and nondecreasing on* $[0, 1/2]$*, then*

$$d_1 \le d_2 \implies \mathbb{E}H(P_{d_1}) \ge \mathbb{E}H(P_{d_2}). \tag{3.1}$$

*If* $H$ *is strictly increasing on* $(0, 1/2)$ *and* $d_1 < d_2$*, then the inequality is strict.*

*Proof.* The case $d_1 = d_2$ is immediate, so assume $d_1 < d_2$. Put $Q_d = P_d(1 - P_d) \in (0, 1/4)$. For

$$p_\pm(q) = \frac{1 \pm \sqrt{1 - 4q}}{2}, \qquad \ell(q) = \log \frac{p_+(q)}{p_-(q)},$$

the density of $Q_d$ is

$$g_d(q) = \frac{2}{B(s/2 + d, s/2 - d)}\, q^{s/2-1}(1 - 4q)^{-1/2} \cosh\bigl(d\ell(q)\bigr).$$

The densities $g_{d_1}$ and $g_{d_2}$ share the factor $q^{s/2-1}(1 - 4q)^{-1/2}$, so

$$\frac{g_{d_2}(q)}{g_{d_1}(q)} = \frac{B(s/2 + d_1, s/2 - d_1)}{B(s/2 + d_2, s/2 - d_2)} \frac{\cosh(d_2\ell(q))}{\cosh(d_1\ell(q))}.$$

The function $\ell$ is strictly decreasing, since $p_+(q)/p_-(q)$ decreases to 1 as $q \uparrow 1/4$. For $d_2 > d_1$, $\cosh(d_2\ell)/\cosh(d_1\ell)$ is strictly increasing in $\ell > 0$, because $d \mapsto d \tanh(d\ell)$ is strictly increasing; hence $g_{d_2}/g_{d_1}$ is strictly decreasing in $q$. Thus $Q_{d_2}$ is smaller than $Q_{d_1}$ in the likelihood-ratio order, and therefore in the usual stochastic order [11, Thm. 1.C.1]. Since both densities integrate to one and their ratio is strictly monotone, it crosses 1 exactly once; hence $g_{d_1} - g_{d_2}$ has a single sign change, negative near 0 and positive near $1/4$, so for every nondecreasing $\phi$,

$$\mathbb{E}\phi(Q_{d_1}) - \mathbb{E}\phi(Q_{d_2}) = \int_0^{1/4} \bigl(\phi(q) - \phi(q^*)\bigr)\bigl(g_{d_1}(q) - g_{d_2}(q)\bigr)\, \mathrm{d}q \ge 0,$$

where $q^*$ is the crossing point and the integrand is nonnegative throughout. Finally, symmetry of $H$ gives $H(p) = \widetilde{H}(p(1-p))$, with $\widetilde{H}$ nondecreasing on $[0, 1/4]$ precisely when $H$ is nondecreasing on $[0, 1/2]$; taking $\phi = \widetilde{H}$ proves (3.1). If $H$ is strictly increasing and $d_1 < d_2$, the likelihood ratio is nonconstant, the crossing is nondegenerate, and the inequality is strict. □

## 4. The cubic residue kernel

For symmetric nonnegative weights $w_k = w_{m-k}$, the kernel of (2.5) is

$$G_{m,w}(p) = \sum_{k=1}^{m-1} w_k \binom{3m-2}{3k-1} p^{3k}(1 - p)^{3(m-k)}, \qquad G_{m,w}(p) = G_{m,w}(1 - p).$$

This section proves that $G_{m,w}$ is monotone on $[0, 1/2]$ whenever the weights increase toward the center.

**Theorem 4.1** (Weighted cubic residue monotonicity)**.** *Suppose*

$$0 \le w_1 \le w_2 \le \cdots \le w_{\lfloor m/2 \rfloor}, \qquad w_k = w_{m-k}. \tag{4.1}$$

*Then* $G_{m,w}$ *is nondecreasing on* $[0, 1/2]$*. If the weights are not all zero, then* $G_{m,w}$ *is strictly increasing on* $[0, 1/2]$*.*

4.1. **The constant-weight kernel.** Let $w_k \equiv 1$, and write $G_m = G_{m,1}$. Set $n = 3m-2$ and $t = p/(1-p)$. For $0 \le p \le 1/2$ one has $0 \le t \le 1$, and

$$G_m\left(\frac{t}{1+t}\right) = \frac{1}{(1+t)^{3m}} \sum_{k=1}^{m-1} \binom{n}{3k-1} t^{3k}.$$

Differentiation gives

$$\frac{\mathrm{d}}{\mathrm{d}t} G_m\left(\frac{t}{1+t}\right) = \frac{3J_m(t)}{(1+t)^{3m+1}}, \tag{4.2}$$

where

$$J_m(t) = \sum_{k=1}^{m-1} \binom{n}{3k-1} t^{3k-1}\bigl(k-(m-k)t\bigr). \tag{4.3}$$

**Lemma 4.2.** *For every $m \ge 2$,*

$$J_m(t) > 0 \qquad (0 < t < 1).$$

*Consequently, $G_m$ is strictly increasing on* $[0, 1/2]$.

*Proof.* Writing $\nu = 3k-1$ in (4.3) gives

$$J_m(t) = \frac{1}{3} \sum_{\substack{0 \le \nu \le n \\ \nu \equiv 2\,(3)}} \binom{n}{\nu} t^{\nu}\bigl((\nu+1) - (n-\nu+1)t\bigr).$$

Introduce the projective Bernstein transform

$$\mathcal{P}_m(\xi) = (1+\xi)^{n+1} J_m\left(\frac{\xi}{1+\xi}\right). \tag{4.4}$$

Then

$$\mathcal{P}_m(\xi) = \frac{1}{3} \sum_{\nu \equiv 2\,(3)} \binom{n}{\nu} \xi^{\nu}(1+\xi)^{n-\nu}\bigl((\nu+1) + (2\nu-n)\xi\bigr). \tag{4.5}$$

For $0 \le j \le n+1$, coefficient extraction from (4.5) gives

$$\begin{aligned}
[\xi^j]\mathcal{P}_m(\xi) &= \frac{1}{3} \sum_{\substack{0 \le \nu \le j \\ \nu \equiv 2\,(3)}} \binom{n}{\nu} \left[(\nu+1)\binom{n-\nu}{j-\nu} + (2\nu-n)\binom{n-\nu}{j-\nu-1}\right] \\
&= \frac{\binom{n+1}{j}}{3(n+1)} S_{n,j}, \qquad (4.6)
\end{aligned}$$

where

$$S_{n,j} = \sum_{\substack{0 \le \nu \le j \\ \nu \equiv 2\,(3)}} \binom{j}{\nu} (n-\nu+1)(2\nu+1-j). \tag{4.7}$$

The second equality in (4.6) follows from the elementary identities

$$\binom{n}{\nu}\binom{n-\nu}{j-\nu} = \binom{n}{j}\binom{j}{\nu}, \qquad \binom{n}{\nu}\binom{n-\nu}{j-\nu-1} = \binom{n+1}{j}\binom{j}{\nu}\frac{j-\nu}{n+1}.$$

The transform is Bernstein-theoretic: writing $J_m(t) = \sum_j b_j \binom{n+1}{j} t^j (1-t)^{n+1-j}$ in the degree-$(n+1)$ Bernstein basis, one has $[\xi^j]\mathcal{P}_m(\xi) = b_j\binom{n+1}{j}$, so nonnegativity of the Bernstein coefficients $b_j$ certifies $J_m(t) \ge 0$ on $[0,1]$ [10]. We claim that every $S_{n,j}$ is nonnegative.

Put $\delta = n+1-j$. A third-root-of-unity filter evaluates the sum over the class $\nu \equiv 2 \pmod 3$; its trigonometric terms are periodic modulo 6 in $j$, so the closed form splits into six residue classes,

$$3S_{n,j} = \begin{cases} \delta(2^j - 3j - 1) - 3j(j-1), & j \equiv 0 \pmod 6, \\ \delta(2^j - 2) - 3j(j-1), & j \equiv 1 \pmod 6, \\ \delta(2^j + 3j - 1), & j \equiv 2 \pmod 6, \\ \delta(2^j + 3j + 1) + 3j(j-1), & j \equiv 3 \pmod 6, \\ \delta(2^j + 2) + 3j(j-1), & j \equiv 4 \pmod 6, \\ \delta(2^j - 3j + 1), & j \equiv 5 \pmod 6. \end{cases} \tag{4.8}$$

In detail, for

$$R_a(j) = \sum_{\nu \equiv a\,(3)} \binom{j}{\nu}$$

and $\omega = e^{2\pi i/3}$,

$$R_a(j) = \frac{1}{3}\sum_{\ell=0}^{2} \omega^{-a\ell}(1+\omega^\ell)^j = \frac{1}{3}\left(2^j + 2\cos\frac{(j-2a)\pi}{3}\right),$$

where $2\cos((j-2a)\pi/3)$ equals $2, 1, -1, -2, -1, 1$ according as $j - 2a \equiv 0, 1, 2, 3, 4, 5 \pmod 6$. These values also follow with no appeal to $\mathbb{C}$: Pascal's rule gives $R_a(j+1) = R_a(j) + R_{a-1}(j)$ (subscripts modulo 3), and this recurrence with $R_0(0) = 1$, $R_1(0) = R_2(0) = 0$ fixes them by induction. The first two moments over the residue class are

$$\sum_{\nu \equiv 2\,(3)} \nu \binom{j}{\nu} = jR_1(j-1), \qquad \sum_{\nu \equiv 2\,(3)} \nu^2 \binom{j}{\nu} = j(j-1)R_0(j-2) + jR_1(j-1).$$

Expanding the summand in (4.7) gives, for $j \geq 2$,

$$\begin{aligned} S_{n,j} &= \delta\bigl(2jR_1(j-1) + (1-j)R_2(j)\bigr) \\ &\quad + j(j-1)\bigl(-2R_0(j-2) + 3R_1(j-1) - R_2(j)\bigr), \end{aligned} \tag{4.9}$$

with $j = 0, 1$ following directly from (4.7). Substituting the values of $R_2(j)$, $R_1(j-1)$, and $R_0(j-2)$ by residue $j \bmod 6$ yields (4.8).

It remains only to inspect the six cases. The cases $j \equiv 2, 3, 4 \pmod 6$ are immediate. If $j \equiv 5 \pmod 6$, then $j \geq 5$ and $2^j - 3j + 1 > 0$. For the two remaining classes, note that $n + 1 = 3m - 1$ is congruent to 2 or 5 modulo 6; hence $\delta \geq 1$ when $j \equiv 1 \pmod 6$, and $\delta \geq 2$ when $j \equiv 0 \pmod 6$ and $j > 0$. If $j \equiv 1 \pmod 6$, the case $j = 1$ gives zero, while for $j \geq 7$,

$$2^j - 2 \geq 3j(j-1). \tag{4.10}$$

Finally, when $j \equiv 0 \pmod 6$ and $j > 0$, one has $j \geq 6$ and

$$2(2^j - 3j - 1) \geq 3j(j-1). \tag{4.11}$$

The inequalities (4.10) and (4.11) hold with equality at $j = 7$ and $j = 6$, respectively. Advancing $j$ by one increases the left-minus-right difference by $2^j - 6j$ in (4.10) and by $2^{j+1} - 6j - 6$ in (4.11), both positive for $j \geq 6$; the inequalities therefore persist for all larger $j$. Likewise $2^j - 3j + 1 > 0$ at $j = 5$, with increment $2^j - 3 > 0$ thereafter. Hence $S_{n,j} \geq 0$ in every case.

By (4.6), all coefficients of $\mathcal{P}_m$ are nonnegative. The coefficient at $\xi^2$ is positive, since $\delta = n - 1 > 0$ and $3S_{n,2} = 9\delta$. Therefore $\mathcal{P}_m(\xi) > 0$ for $\xi > 0$. The substitution $t = \xi/(1+\xi)$ in (4.4) proves $J_m(t) > 0$ for $0 < t < 1$, and (4.2) completes the proof. $\square$

4.2. **Insertion of monotone weights.** Arbitrary weights satisfying (4.1) preserve nonnegativity of the derivative.

For $0 < t < 1$, differentiation of (2.5) gives

$$\frac{\mathrm{d}}{\mathrm{d}t} G_{m,w}\left(\frac{t}{1+t}\right) = \frac{3J_{m,w}(t)}{(1+t)^{3m+1}}, \tag{4.12}$$

where

$$J_{m,w}(t) = \sum_{k=1}^{m-1} w_k \binom{3m-2}{3k-1} t^{3k-1}\big(k-(m-k)t\big). \tag{4.13}$$

Pair the terms indexed by $k$ and $m-k$. For $1 \le k < m/2$, let

$$\begin{aligned} B_{m,k}(t) = \binom{3m-2}{3k-1} \Big[ & t^{3k-1}\big(k-(m-k)t\big) \\ & + t^{3(m-k)-1}\big((m-k)-kt\big)\Big]. \end{aligned} \tag{4.14}$$

If $m$ is even, the central block is

$$B_{m,m/2}(t) = \binom{3m-2}{3m/2-1} \frac{m}{2} t^{3m/2-1}(1-t) > 0. \tag{4.15}$$

Then

$$J_{m,w}(t) = \sum_{k=1}^{\lfloor m/2 \rfloor} w_k B_{m,k}(t),$$

with the convention (4.15) at the center.

**Lemma 4.3.** *For each fixed $m \ge 2$ and $0 < t < 1$, the sequence*

$$B_{m,1}(t), B_{m,2}(t), \ldots, B_{m,\lfloor m/2 \rfloor}(t)$$

*has no more than one sign change, from negative to positive, and its final term $B_{m,\lfloor m/2 \rfloor}(t)$ is positive.*

*Proof.* For $k < m/2$, the sign of $B_{m,k}(t)$ is the sign of

$$H_{m,k}(t) = k-(m-k)t + t^{3(m-2k)}\big((m-k)-kt\big).$$

Put $d = m-2k > 0$ and $t = e^{-z}$, $z > 0$. Direct rearrangement gives

$$\begin{aligned} H_{m,k}(t) &= \frac{1}{2}(1+t)(1+t^{3d}) \left[ m\frac{1-t}{1+t} - d\frac{1-t^{3d}}{1+t^{3d}} \right] \\ &= \frac{1}{2}(1+t)(1+t^{3d}) \left[ m \tanh\frac{z}{2} - d \tanh\frac{3dz}{2} \right]. \end{aligned} \tag{4.16}$$

The function $d \mapsto d \tanh(3dz/2)$ is strictly increasing for $d > 0$. Therefore, for $2 \le k < m/2$, if $B_{m,k}(t) \le 0$, then the same bracket in (4.16) is strictly negative after replacing $d$ by $d+2$, which corresponds to replacing $k$ by $k-1$. Thus

$$B_{m,k}(t) \le 0 \quad \Longrightarrow \quad B_{m,k-1}(t) < 0 \qquad (2 \le k < m/2).$$

The final block is always positive. For even $m$ this is (4.15). For odd $m$, the final block has $d = 1$, and its bracket in (4.16) is $m\tanh(z/2) - \tanh(3z/2) > 0$, because $m \ge 3$ and $\tanh(3y) < 3\tanh y$ for $y > 0$. Hence the blocks have the asserted sign pattern. $\square$

*Proof of Theorem 4.1.* By Lemma 4.2,

$$\sum_{k=1}^{\lfloor m/2 \rfloor} B_{m,k}(t) = J_m(t) > 0 \qquad (0 < t < 1).$$

The block sequence has one sign change by Lemma 4.3. Applying Lemma 3.1 with the nondecreasing weights in (4.1) gives

$$J_{m,w}(t) = \sum_{k=1}^{\lfloor m/2 \rfloor} w_k B_{m,k}(t) \geq 0.$$

Equation (4.12) proves that $G_{m,w}$ is nondecreasing on $[0, 1/2]$.

Suppose now that the weights are not all zero, and put $L = \lfloor m/2 \rfloor$. Then $w_L > 0$, and $B_{m,L}(t) > 0$ by Lemma 4.3. Since $\sum_{k=1}^{L} B_{m,k}(t) = J_m(t) > 0$, the strict part of Lemma 3.1 gives $J_{m,w}(t) > 0$ for $0 < t < 1$, and $G_{m,w}$ is strictly increasing on $[0, 1/2]$. □

## 5. Proof of the theorem

The representation (2.4) and the two monotonicity facts of Sections 3 and 4 combine to show that the coefficient convolution is Schur-concave in its two parameters [8, Def. 3.A.1], which gives Theorem 1.1.

**Proposition 5.1.** *For each $m \geq 2$ and each log-concave sequence $(f_n)$ as in Theorem 1.1, the symmetric function $(u, v) \mapsto C_{m,f}(u, v)$ is Schur-concave on $(0, \infty)^2$. Equivalently, at fixed $s = u + v$,*

$$d \longmapsto C_{m,f}\left(\frac{s}{2} + d, \frac{s}{2} - d\right)$$

*is nonincreasing for $0 \leq d < s/2$. If at least one product $f_k f_{m-k}$ is positive, then this function is strictly decreasing.*

*Proof.* The prefactor in (2.4) depends only on $s = u + v$, and $G_{m,w}$ is symmetric by (2.5). By Theorem 4.1 it is nondecreasing on $[0, 1/2]$, and strictly increasing when at least one weight $w_k = f_k f_{m-k}$ is positive; Lemma 3.2 and its strict part give the two conclusions. □

*Proof of Theorem 1.1.* The coefficients of degrees 0 and 1 in (1.3) vanish. For $m \geq 2$, direct multiplication gives

$$\begin{aligned} [x^m]\Delta_f(\mu; \alpha, \beta; x) &= C_{m,f}(\mu + \alpha, \mu + \beta) \\ &\quad - C_{m,f}(\mu, \mu + \alpha + \beta). \end{aligned} \tag{5.1}$$

Assume first that $\mu > 0$ and $\alpha, \beta > 0$. The two parameter pairs in (5.1) have the same sum $s = 2\mu + \alpha + \beta$ and imbalances

$$d_1 = \frac{|\alpha - \beta|}{2} \leq \frac{\alpha + \beta}{2} = d_2.$$

Proposition 5.1 therefore gives

$$C_{m,f}(\mu + \alpha, \mu + \beta) \geq C_{m,f}(\mu, \mu + \alpha + \beta).$$

This proves nonnegativity of every coefficient. If $\alpha = 0$ or $\beta = 0$, the Turánian is identically zero. The boundary case $\mu = 0$ follows from $F_f(0; x) = 0$ and the nonnegative coefficients of $F_f(\alpha; x)F_f(\beta; x)$. □

**Corollary 5.2.** *Under the hypotheses of Theorem 1.1, let $\mu \geq 0$, $\alpha, \beta > 0$, and $m \geq 2$. Then*

$$[x^m]\Delta_f(\mu; \alpha, \beta; x) > 0$$

*if and only if there exists $1 \leq k \leq m - 1$ with $f_k f_{m-k} > 0$.*

*Proof.* If every product $f_k f_{m-k}$ vanishes, then both terms of (5.1) vanish. Conversely, suppose one such product is positive. If $\mu > 0$, the two imbalances in the proof of Theorem 1.1 satisfy $d_1 < d_2$, so the strict part of Proposition 5.1 gives strict inequality. If $\mu = 0$, then $F_f(0; x) = 0$, so the coefficient equals $C_{m,f}(\alpha, \beta)$, a sum of nonnegative terms with at least one strictly positive. □

**Corollary 5.3.** *Suppose $(f_n)$ is not identically zero, and let $R$ be the radius of convergence of $\sum_{n \geq 1} f_n x^n$. For every $0 < x < R$, the function $\mu \mapsto F_f(\mu; x)$ is finite, continuous, and positive on $(0, \infty)$, and $\mu \mapsto \log F_f(\mu; x)$ is concave there.*

*Proof.* Log-concavity with no internal zeros makes $f_{n+1}/f_n$ nonincreasing on the support of $(f_n)$, so $R > 0$. For each fixed $\mu > 0$, Stirling's formula gives

$$\frac{(\mu)_{3n}}{(3n-1)!} = \frac{\Gamma(3n+\mu)}{\Gamma(\mu)\Gamma(3n)} \sim \frac{(3n)^\mu}{\Gamma(\mu)} \qquad (n \to \infty).$$

Hence $F_f(\mu; \cdot)$ has the same radius of convergence $R$ as $\sum_{n\geq 1} f_n x^n$. On every compact set of positive $\mu$ and for $0 < x < R$, the series converges locally uniformly in $\mu$, since its extra factor grows at most polynomially in $n$; thus $\mu \mapsto F_f(\mu; x)$ is continuous. Since $(f_n)$ is not identically zero, its nonnegative coefficients give $F_f(\mu; x) > 0$ for $\mu, x > 0$. By Theorem 1.1,

$$F_f(\mu+\alpha; x)F_f(\mu+\beta; x) \geq F_f(\mu; x)F_f(\mu+\alpha+\beta; x)$$

for all nonnegative shifts. Taking $\alpha = \beta = h > 0$ gives $F_f(\mu+h; x)^2 \geq F_f(\mu; x)F_f(\mu+2h; x)$, so $\mu \mapsto \log F_f(\mu; x)$ is midpoint concave, and by continuity concave, on $(0, \infty)$. $\square$

**Remark 5.4.** The interval-support condition cannot be dropped. Put $f_1 = f_4 = 1$ and $f_n = 0$ otherwise. The two positive indices differ by 3, so $f_{n-1}f_{n+1} = 0$ at every index where the three terms are defined and $f_n^2 \geq f_{n-1}f_{n+1}$ holds throughout, strictly at $n = 4$. The positive support $\{1, 4\}$ is not an interval. A single zero between two positive terms violates the inequality at that zero, so a gap of two is the shortest the inequality admits. At $m = 5$ the weights $w_k = f_k f_{5-k}$ are $w_1 = w_4 = 1$ and $w_2 = w_3 = 0$, decreasing toward the center rather than increasing, and

$$[x^5]\Delta_f(2; 1, 1; x) = -9360.$$

## 6. Concluding remarks

The proof of Theorem 1.1 hinges on the special feature of the residue class modulo 3. After the beta-binomial reduction, the entire conjecture is controlled by the monotonicity of

$$p(1-p)\mathbb{P}\{\mathrm{Bin}(3m-2, p) \equiv 2 \pmod 3\}.$$

The projective transform (4.4) converts its derivative into a polynomial whose Bernstein coefficients are nonnegative. This positivity is not termwise obvious in the monomial basis; it appears only after the root-of-unity filter and its mod-6 case split. Log-concavity enters the proof of Theorem 1.1 exactly once: it makes the symmetric weights $w_k = f_k f_{m-k}$ nondecreasing as $k$ approaches $m/2$.

For a general integer $r$, the same beta-binomial reduction produces, in the constant-weight case,

$$p(1-p)\mathbb{P}\{\mathrm{Bin}(rm-2, p) \equiv r-1 \pmod r\},$$

and in general its analogue weighted by the products $f_k f_{m-k}$. The cubic proof suggests testing whether a suitable projective degree lift has a positive coefficient array. The known failure at $r = 4$ shows that such positivity cannot persist without an additional restriction; locating the first coefficient obstruction in the lifted array may explain that failure.

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